

Defocus in Match Theory: Remarks on Yip & Yuan 2025

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1 Introduction

Yip & Yuan (2025) add a new case to the literature on syntax-prosody mismatches, arguing that in Cantonese and Mandarin, a defocused phrase within a right-dislocated CP is prosodically grouped with material from a constituent to its left: roughly, $[_{CP} XP YP] [_{CP} ZP_{defoc}] \rightarrow (XP) (YP ZP_{defoc})$. They present an analysis based in Match Theory (Selkirk 2011), proposing a new constraint DEFOCUS which, together with several other prosodic constraints, is argued to outrank MATCH(CP,t) and produce the mismatch.

I show that sharpening and clarifying these lines of analysis is necessary to advance inquiry on defocus-driven mismatches, especially in light of recent work in syntax-prosody mapping in OT which emphasizes the necessity of taking all candidates and all constraint violations into account (Bellik & Kalivoda 2016, 2019; Kalivoda 2018, 2022; Van Handel 2021; Bellik, Ito, Kalivoda & Mester 2022; Ishihara & Kalivoda 2022; Alber & Prince 2022; Elkins 2023; and papers in Bellik, Ito, Kalivoda & Mester 2023). The overall research program is based on developments in modern rigorous OT demonstrating that assertions of optimality are only meaningful within well-defined *OT systems* (Bane & Riggle 2012; Alber, DelBusso & Prince 2016; Prince 2017a,b,c). Such analyses are facilitated by computational tools like SPOT, developed specifically for candidate generation and constraint evaluation in syntax-prosody analyses, and OTWorkplace, which calculates OT grammars and typologies (Bellik, Bellik & Kalivoda 2015–2023; Prince, Merchant & Tesar 2007–2025).

Yip & Yuan’s OT analysis is intriguing, but incomplete due to the omission of important candidates. Basing an OT system on their constraints and the candidates included in their tableaux (§2.1), I show that one desired phrasing ties with three unwanted competitors, while another desired phrasing ties with another (§2.2). Thus, to achieve its goals, the analysis needs to be sharpened to exclude candidates that are unavoidable when the formal system is developed. I then show in §2.3 that adding two more constraints to the system, $*_t$ and BINMAX(φ ,branches), successfully harmonically bounds the unwanted competitors (§2.3). However, there is an additional issue regarding input-output correspondence relations and the constraint DEFOCUS, avoided by stipulation in §2 and by omission of candidates in Yip & Yuan’s paper, that I take up in §3. There, I show that if a certain assumption about input-output correspondence relations is not added to Yip & Yuan’s implicit theory of GEN, and if prosodic headedness is not taken into account at the utterance level, a perfectly matching candidate harmonically bounds the desired optima. I construct an additional

OT system, further refining assumptions of GEN and CON, and show that this problem can also be solved.

2 Co-optimality: discovered and resolved

Prince (2017c) shows that because optimality in OT means being better than *all* competitors according to some ranking of *all* constraints, it is only well-defined within an *OT system*. An OT system S is a pair $\langle S.GEN, S.CON \rangle$. $S.GEN$ defines the candidate sets for S (in the case of syntax-prosody mapping, pairs of syntactic inputs and prosodic outputs), while $S.CON$ defines the constraints of S as functions from candidates to non-negative integers.

While systems $S = \langle S.GEN, S.CON \rangle$ are not always spelled out in full detail, they have been at least implicit in all work in OT since its inception, and an author’s intentions are often clear enough to allow the details of the implicit system to be determined by careful inspection of tableaux, constraint definitions (even when these fail to explain violation-counting), and general knowledge of common assumptions in the field. While Yip & Yuan (2025) do not explicitly lay out an OT system, their proposals are clear on most points and permit the study of a system that—as far as I can tell—is entirely compatible with their discussion and tableaux.

In the following subsection (§2.1), I define a system DRP (‘Defocus Rephrasing’) based as closely as possible on what I take Yip & Yuan’s proposals to be. In §2.2, I show that the factorial typology of DRP does not contain unique desired winners, but that there are unwanted co-optima. Then in §2.3, I show how the co-optimality problem can be resolved in an expanded system $DRP+$.

2.1 The System DRP

In terms of output representations, DRP is situated within Prosodic Hierarchy Theory (Selkirk 1984; Nespor & Vogel 1986), in particular within a version of the theory with partial Weak Layering (Ito & Mester 1992/2003) and prosodic hierarchy $\upsilon > \iota > \varphi > \omega$.¹ As is standard, υ represents the utterance, ι the intonational phrase, φ the phonological phrase, and ω the prosodic word.

To begin defining a system DRP based on Yip & Yuan’s OT analysis, we must first define $DRP.GEN$, which specifies the sole candidate set of DRP and the candidates it contains. As in most work on syntax-prosody mapping, the input is a syntactic tree and outputs are prosodic trees.

- (1) $DRP.GEN$ (based on candidates found in Yip & Yuan’s 2025 tableaux)²

- a. *Input*: $[[_{CP} [_{XP} X] [_{YP} Y]] [_{CP} [_{ZP} Z]_{defoc}]$

where the whole structure is a linearized syntactic tree rooted in an unlabeled con-

¹Feet and syllables below the ω level play no role in the discussion to follow, so $DRP.GEN$ treats ω ’s as atomic.

²The candidate sets for $DRP.GEN$ and the other systems discussed below were generated using a combination of SPOT and some post-processing. First, SPOT was used to get all potential subtree children of υ , without headedness marked. New versions of each of these subtrees were then constructed with every licit combination of (un)headedness on nodes. The resulting subtrees were cross-concatenated to yield 180 complete output representations, listed in Appendix B. Cross-concatenation and constraint violation counting were done with JavaScript. The resulting violation tableaux were then imported into OTWorkplace for typology calculation. The generation and evaluation process was not done entirely in SPOT since the necessary representations are too complex given SPOT’s current capabilities, with four (rather than three) prosodic categories and headedness at multiple levels.

stituent,³ with syntactic words X, Y, Z, syntactic phrases XP, YP, ZP, and clauses CP, with subscript “defoc” suffixed to a constituent bearing a [defocus] feature

b. *Outputs*: An output is any prosodic tree P for which the following conditions hold:

- i. *v-Rootedness*
The root node of P is of category v .
- ii. *v-Contains- ι*
At least one child of the root v is of category ι .
- iii. *v-Children*
Every child of the root v is of category φ or ι .
- iv. *v-Unheadedness*
The root v is unheaded, meaning that none of its children is marked as a head.
- v. *ι -Children*
All children of ι nodes are of category φ .
- vi. *Optional ι -Headedness*
Zero or one child of every ι node is marked as that node’s head.
- vii. *φ -Children*
The children of a φ node are either all of category φ or all of category ω .
- viii. *No Vacuous Recursion*
If a φ is unary-branching, its sole child is of category ω .
- ix. *φ -Headedness*
Exactly one child of every φ node is marked as that node’s head.
- x. *Terminal Correspondence*
P has exactly three terminal nodes, and these are of category ω . They are ω_X , ω_Y , ω_Z , corresponding to the input words X, Y, Z, respectively.
- xi. *Terminal Order*
The linear order of P’s terminal nodes is $\omega_X \omega_Y \omega_Z$.

The input contains three syntactic words: X, Y, Z. For ease of reference, their labels differ from Yip & Yuan’s (Z, Y, and Z), but this is purely notational.⁴

Yip & Yuan (2025) do not in fact make clear whether the input they are considering in their OT analysis contains three words or simply three phrases of unspecified length. The Cantonese and Mandarin examples given prior to the OT analysis contain more than three words, but the discussion of the OT analysis seems to suggest that they are considering three words in their tableaux. Inputs with more than three words could of course be incorporated into a larger system based on DRP. Even if looking at a three-word sentence in DRP is an abstraction (since we are ultimately interested in an analysis that works for longer sentences as well), doing an OT analysis requires making a concrete decision on this point, and a three-word input is a simple place to start.

³The root node, here unlabeled, is not shown in Yip & Yuan’s inputs but is presumably present. I call it “unlabeled” because, whatever its categorial nature, Yip & Yuan do not treat it as a CP or XP visible to the mapping constraints. An alternative would be to take the input to be a graph-theoretic forest (dropping the outer brackets), but with a linear order imposed on its two connected components (the CPs).

⁴Note that the letter ‘X’ is doing double-duty, since in the constraint name MATCH(XP, φ) introduced below, ‘XP’ refers to any non-CP phrase (here XP, YP, or ZP) and not just to the one dominating the first word of the input.

The output condition *v-Rootedness* asserts that every output is a single utterance v . By the conditions *v-Contains- ι* and *v-Children*, the utterance immediately dominates at least one intonational phrase ι and zero or more phonological phrases φ . Although prosodic heads play a major role in Yip & Yuan’s (2025) analysis, utterances are not shown as headed in their tableaux, so in DRP.GEN, no node immediately dominated by v is designated as a prosodic head (*v-Unheadedness*), a point to which I return in §3.⁵

The output conditions *ι -Children* and *Optional ι -Headedness* deal with intonational phrases ι and their children, all of which are phonological phrases φ . An ι can lack a head or have exactly one. In the text, but not in the candidates in their tableaux, Yip & Yuan (2025) consider the possibility of multiply-headed constituents. While this is an intriguing idea, it is not necessary to allow such constituents in DRP.GEN, as such structures are not claimed to be instantiated in Cantonese or Mandarin. Yip & Yuan also entertain the possibility of recursive intonational phrases, but only show such candidates in one tableau (their (98)), dismissing them as a possibility for Cantonese and Mandarin defocus phrasing. I therefore also exclude candidates with ι -recursion from DRP.GEN as an analytical simplification, though an expanded system could easily be defined to include such candidates.

The output condition *φ -Children* states that every phonological phrase dominates children that all share a prosodic category, which is either φ or ω . This means that EQUALSISTERS, which Myrberg (2013) introduces as a violable constraint, is enforced in DRP.GEN as an inviolable condition specifically for arrays of children of phonological phrases.⁶ That is, while DRP.GEN admits phonological phrases of the form $(\varphi \varphi \varphi)$, it does not admit phonological phrases with children of distinct categories like $(\varphi \omega \varphi)$, even though such structures are allowed in typical syntax-prosody systems (Bellik, Ito, Kalivoda & Mester 2023). The reason for this simplification is that Yip & Yuan (2025) do not include any candidates violating *φ -Children* in their tableaux, so setting aside the complication of phrases like $(\varphi \omega \varphi)$ suffices for the present discussion (and cuts down on the size of the candidate set). As with ι -recursion, absent from DRP but present in other systems, it is possible to define an alternative system based on DRP but without the EQUALSISTERS-like condition in *φ -Children*, possibly introducing EQUALSISTERS as a violable constraint, but here I focus only on DRP.

The *No Vacuous Recursion* condition states that the child of a unary phonological phrase is a prosodic word ω . This rules out phonological phrases of the form $(\varphi \varphi)$, which to my knowledge have never been proposed in any analysis (which is why vacuous recursion is excluded as an option in SPOT; see Bellik & Kalivoda 2019).

The condition *φ -Headedness* states that every phonological phrase φ has a unique designated head (in contrast to the ι , which can lack a head, and the v , which cannot have a head). In Yip & Yuan’s (2025) tableaux, branching φ ’s are always uniquely headed, and the possibility of unheaded φ ’s does not figure into their analysis. By contrast, Yip & Yuan do not indicate headedness

⁵As Yip & Yuan (2025) point out, their notion of prosodic headedness refers to abstract phonological prominence (Elordieta & Selkirk 2018) and thus differs from the notion of headedness from Selkirk (1996), which requires every nonterminal node to immediately dominate at least one node of the next lowest category on the prosodic hierarchy. Following Yip & Yuan (2025), DRP respects this latter condition while allowing *prominence*-unheaded intonational phrases.

⁶No EQUALSISTERS-like condition holds in DRP for the children of the utterance v , which by *v-Children* may immediately dominate a φ . The reason for this difference is simply that Yip & Yuan (2025) show candidates in which the v immediately dominates both ι and φ , but none in which a φ immediately dominates both a φ and an ω .

for unary φ 's, but barring a stipulation to the effect that unary φ 's are obligatorily unheaded while branching φ 's are obligatorily headed, it seems reasonable to suppose that unary φ 's must be headed by their unique children. This point of uncertainty in interpretation of candidate representations underscores the need for a fully defined S.GEN within a system S. In DRP the question is settled by φ -Headedness.

The conditions *Terminal Correspondence* and *Terminal Order* concern the terminal strings of output trees, all of which are three prosodic words ω . The n^{th} terminal node of the input stands in *correspondence* (McCarthy & Prince 1995) with the n^{th} terminal node of the output and vice versa (nothing is deleted, inserted, or re-ordered), such that the input has terminal string X Y Z and the output has terminal string $\omega_X \omega_Y \omega_Z$.

Finally, before moving on to defining DRP.CON, it is necessary to address one superficial difference between Yip & Yuan's (2025) candidates and those of DRP. While DRP recognizes terminal-to-terminal correspondence (see *Terminal Correspondence* and *Terminal Order*), Yip & Yuan have a more advanced notion of nonterminal-nonterminal correspondence (like corresponding XPs and φ 's) as well as nonterminal-terminal correspondence (like corresponding XPs and ω 's). This is indicated by their sporadic use of correspondence indices on nonterminals. (Only "correspondences of special interest" are indicated in this manner, while correspondence relations between most input and output nodes are left unstated.) These correspondence relations are relevant to the evaluation of their constraint DEFOCUS, but enumerating candidates for a system with free correspondence indexation on both terminals and nonterminals is a daunting challenge and would explode the size of the candidate set, potentially to a size that would be unmanageable unless one or more harmonic bounding theorems could be proved.

Luckily, the advanced notion of correspondence employed by Yip & Yuan and necessary for the evaluation of DEFOCUS can be quasi-faithfully reconstrued with exclusive reference to terminal-to-terminal correspondence, since whenever an input nonterminal node stands in correspondence with an output node in Yip & Yuan's tableaux, its output correspondent is the highest non- ι node in the tree that shares a terminal string with the indicated input correspondent. Terminal-to-terminal correspondence therefore suffices to reproduce the analysis in DRP, in a way proposed by Ito, Kitagawa & Mester (1996): if two constituents α , β have the same terminal string (i.e. if every terminal in α corresponds to a terminal in β and vice versa), then α and β stand in nonterminal-nonterminal correspondence (cf. the definition of *matching* introduced below). While I do not include nonterminal-nonterminal correspondence along the lines of Ito, Kitagawa & Mester (1996) in DRP.GEN, the idea will be baked into the definition of DEFOCUS in DRP.CON.

The constraint set DRP.CON is faithfully based on the constraints employed by Yip & Yuan (2025), but reworded and with name styling slightly adjusted. In cases where constraint definitions did not specify how to count violations, the definitions given here are tailored to produce the same violation counts found throughout Yip & Yuan's tableaux.

(2) DRP.CON (from Yip & Yuan 2025, reworded)

- a. MATCH(CP, ι) (M.CP)
Violated once by every input CP for which there is no matching output ι . (See (3) for the definition of "matching".)
- b. MATCH(XP, φ) (M.XP)
Violated once by every input XP for which there is no matching output φ . (An 'XP' refers to any non-CP syntactic phrase, here including those labeled XP, YP, ZP. See (3))

for the definition of “matching”.)

- c. PARSE (PRS)
Violated once by every φ that is immediately dominated by v .
- d. ι HEAD (ι H)
Violated once by every ι that does not immediately dominate a node designated as a prosodic head.
- e. ALIGN- ι H-R (A ι HR)
Violated once by every prosodic head immediately dominated by an ι within which it is non-rightmost.
- f. DEFOCUS (DF)
Violated once if the highest non- ι node (φ or ω) in the output that shares a terminal string with a [defoc] input constituent, is designated as a prosodic head.
- g. NONRECURSIVITY (NONREC, NR)
Violated once by every node of category κ that dominates another node of category κ .

The constraints in (2a–b) are standard MATCH constraints (Selkirk 2011). While Yip & Yuan (2025) use Selkirk’s original definitions, I follow Elfner (2012, 2015) and many others (see, e.g., Bellik, Ito, Kalivoda & Mester 2023) in defining matching in terms of domination of terminal nodes, as spelled out in (3).

- (3) *Matching* (based on Elfner 2012, 2015; wording from Kalivoda 2022; cf. also Ito, Kitagawa & Mester 1996, pp. 248-49)
Two constituents α , β are matching iff every terminal node in α corresponds to a terminal node in β and every terminal node in β corresponds to a terminal node in α .

Using this definition of matching does not result in any difference in violation counts for MATCH constraints in Yip & Yuan (2025)’s tableaux and in the system DRP.

With the exception of DEFOCUS, which is a new proposal by Yip & Yuan, the other constraints of DRP.CON are either common in the syntax-prosody or broader OT literature or are instantiations of common constraint forms (see Yip & Yuan 2025 for citations). Note that, as Yip & Yuan point out, the alignment constraint ALIGN- ι H-R is categorical, meaning that it is not a function of distance of the ι -head from the right edge, but merely asks whether the ι -head is rightmost within its containing ι . The version of NONRECURSIVITY in (2g) is from Selkirk (1996).⁷

Having defined DRP, I now examine two languages of interest in its factorial typology, showing that the desired outputs for Cantonese and Mandarin tie with unwanted competitors.

2.2 Unwanted ties in DRP

Yip & Yuan (2025) discuss two attested outputs for the input to DRP, both with roughly the same syntax-prosody mismatch, but differing in terms of whether words ω_Y and ω_Z are parsed into unary-branching φ ’s or are sisters within a binary-branching φ . They call speakers with the former phrasing Group 1 and those with the latter Group 2. The specific phrasings are shown in (4).

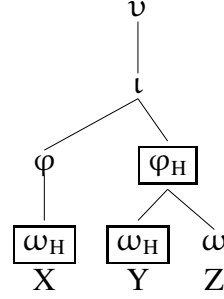
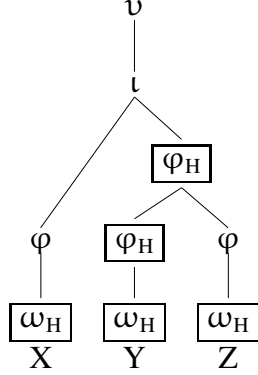
⁷This version of NONRECURSIVITY is what SPOT calls the “parent” version (Bellik et al. 2015–2023). Its definition contrasts with some others found in the literature, notably Truckenbrodt’s (1995, 1999) version that compares terminal strings and a “child-oriented” version (Bellik & Kalivoda 2017).

Prosodic heads are in boxes and suffixed with subscript ‘H’; ‘X’, ‘Y’, and ‘Z’ are written below ω to indicate correspondence.

(4) *Desired outputs* (Yip & Yuan 2025)

a. *Group 1 Phrasing*

b. *Group 2 Phrasing*



For DRP to succeed, its factorial typology needs to contain one language in which the Group 1 phrasing (4a) is the sole optimum, and another in which the Group 2 phrasing (4b) is the sole optimum. Among its six languages (shown in (29) in Appendix A), the typology of DRP does include two languages that *contain* the forms in (4) (in Houghton’s 2012 sense of containment), but not as unique optima. That is, both of the desired outputs tie with unwanted competitors.

This can be ascertained simply by inspecting the violation profiles of the desired outputs and the competitors that tie with them. The Group 1 phrasing is shown in (5a), and the competitors that tie with it in (5b–d). The order of columns here (and elsewhere) is arbitrary and does not indicate any ranking.

(5) *Group 1 desired optimum ties with three other candidates in DRP*

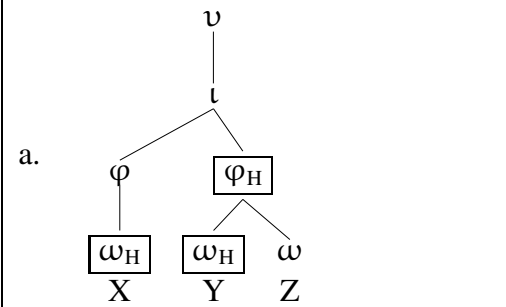
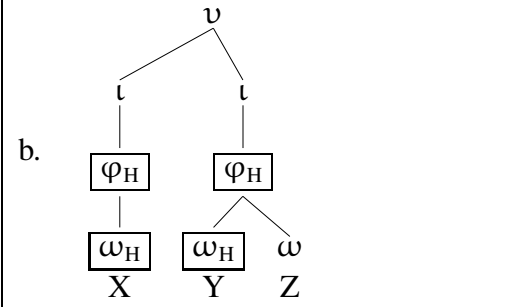
[[_{CP} [_{XP} X] [_{YP} Y]] [_{CP} [_{ZP} Z] _{defoc}]]		M.CP	M.XP	PRS	tH	A _t HR	DF	NR
a.		2	0	0	0	0	0	1

b.	2	0	0	0	0	0	1
c.	2	0	0	0	0	0	1
d.	2	0	0	0	0	0	1

Each of the candidates in (5) incurs two violations of $\text{MATCH}(\text{CP}, \iota)$, one violation of NONRECURSIVITY , and zero violations of the other constraints. Since these candidates have identical violation profiles, they necessarily share a fate within the system, in this case as co-optima within one language in the typology.

The desired Group 2 phrasing (4b) is shown in (6a), and the unwanted candidate with an identical violation profile in (6b).

(6) *Group 2 desired optimum ties with one other candidate in DRP*

[[_{CP} [_{XP} X] [_{YP} Y]] [_{CP} [_{ZP} Z] _{defoc}]]		M.CP	M.XP	PRS	ιH	AιHR	DF	NR
a. 		2	2	0	0	0	0	0
b. 		2	2	0	0	0	0	0

Both (6a) and (6b) incur two violations of $\text{MATCH}(\text{CP}, \iota)$ and two of $\text{MATCH}(\text{XP}, \varphi)$, while satisfying all other constraints. It is thus impossible to distinguish these, violation-wise, within this system.

Within DRP, therefore, the analysis does not succeed; the phrasings in (4) are not unique optima within the factorial typology. To get the right results, a new system is needed, one with constraints that favor these phrasings over their unwanted competitors. In the next subsection, I define a system based on DRP that solves these problems.

2.3 Resolving unwanted ties with DRP+

The unwanted ties in DRP can be resolved by defining a new system, DRP+, with two additional constraints: $\ast\iota$ and $\text{BINMAX}(\varphi, \text{branches})$. This is evident from the fact that the unwanted ties all either contain two ι 's as opposed to one, as in (5b) and (6b), or a ternary-branching φ , as in (5c–d). The full definition of DRP+ is the following:

- (7) $\text{DRP+}.\text{GEN}$
 $= \text{DRP}.\text{GEN} (1)$
- (8) $\text{DRP+}.\text{CON}$
 All constraints of $\text{DRP}.\text{CON} (2)$, plus:
 - a. $\ast\iota$
 Violated once by every ι .
 - b. $\text{BINMAX}(\varphi, \text{branches})$ (BM. φ)
 Violated once by every φ that immediately dominates more than two nodes.

* ι is part of the *STRUC family (Prince & Smolensky 1993/2004), akin to * φ from Truckenbrodt (1995, 1999). BINMAX(φ ,branches) is a commonly used binarity constraint (Elfner 2012; Bellik, Ito, Kalivoda & Mester 2022, among many others). It is important to distinguish it from an alternative notion of φ -binarity counting the number of words dominated by φ (“leaf-counting binarity”, in SPOT terminology; see Bellik & Van Handel 2023 for theoretical discussion and an overview of different types of binarity in the syntax-prosody mapping literature).

The factorial typology of DRP+ contains eight languages as opposed to DRP’s six (all shown in (30) in Appendix A). Among them are two in which the desired phrasings, (4a–b), are the unique optima. The unwanted tied competitors shown in (5–6) are harmonically bounded in DRP+, meaning they do not win under any ranking. This can be seen from the following comparative tableaux (where ‘e’ cells are left blank for visual clarity):⁸

- (9) *Comparative tableau showing harmonic bounding of (5b–d) by (5a) in DRP+*

Winner	Loser	* ι	BM. φ	M.CP	M.XP	PRS	ι H	A ι HR	DF	NR
(5a)	(5b)	W								
(5a)	(5c)		W							
(5a)	(5d)		W							

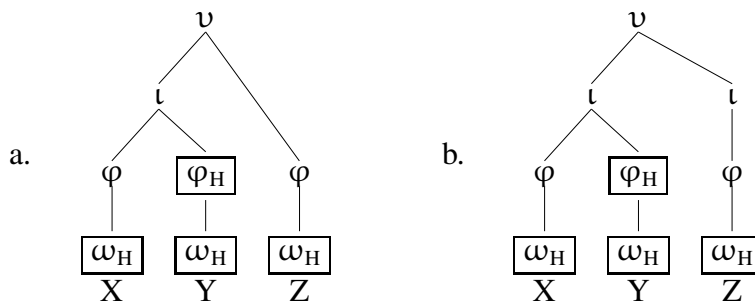
- (10) *Comparative tableau showing harmonic bounding of (6b) by (6a) in DRP+*

Winner	Loser	* ι	BM. φ	M.CP	M.XP	PRS	ι H	A ι HR	DF	NR
(6a)	(6b)		W							

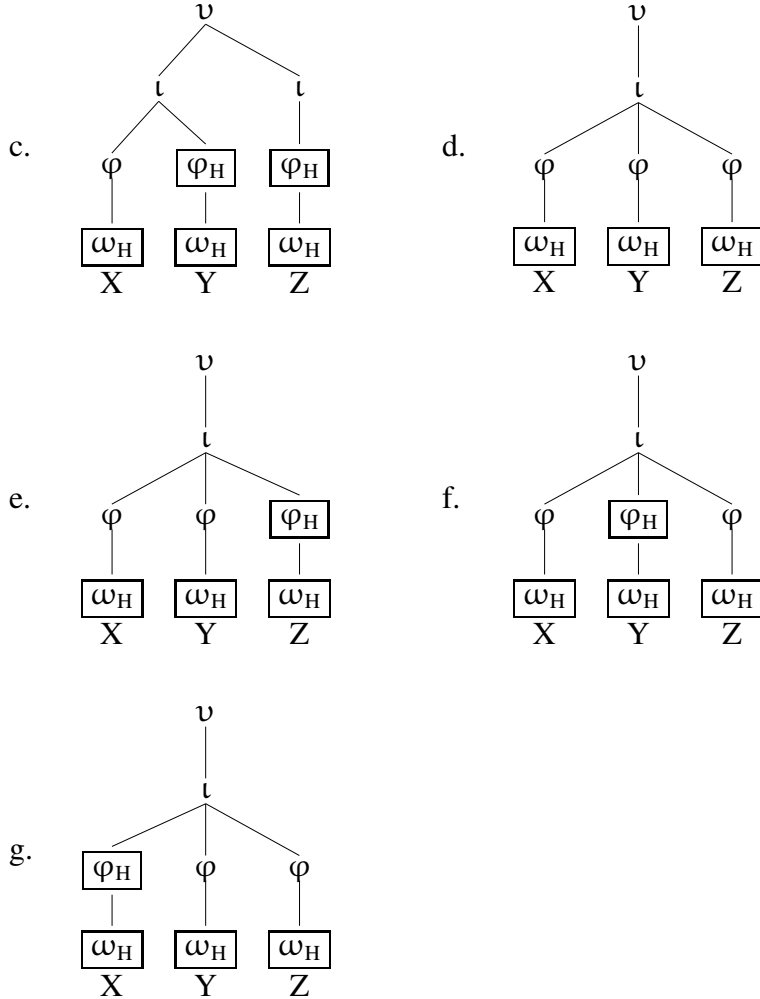
Every row of (9) and (10) shows the desired winner in the winner column and an unwanted competitor in the loser column. Every constraint assesses either ‘W’ or ‘e’ for each comparison, meaning that every constraint either prefers the intended winner or has no preference. The co-optimality problem in DRP is therefore solved in DRP+.

Of course, the comparative tableaux in (9) and (10) do not establish that the desired optima beat all competitors, nor do they establish their grammars. To do this, we need to compare them against all optima within the system. The optima of DRP+ other than (4a–b) are the following:

- (11) *Optima other than (4a–b) in DRP+*



⁸Normally, comparative tableaux should also have an input column. Since there is only one input in the system DRP+, we can exclude this.



Of these, each is sole optimum within a language in the DRP^+ factorial typology, except for (11f–g), which are co-optima within a single language. (Recall that, following Yip & Yuan, AtHR is categorical in both DRP and DRP^+ , and therefore unable to distinguish between degrees of distance from the right edge of l .)

In addition to the eleven optima of DRP^+ in (4) and (11), there are 169 harmonically bounded candidates. These do not provide any information about the *grammars* of the languages of DRP^+ , but the languages of the typology themselves cannot be determined without them, since the question of which candidates allowed by $\text{DRP}^+.\text{GEN}$ are optima and which are harmonically bounded is not known prior to calculation of the typology.

Comparing the desired optima of the two languages (“groups”) in (4) with all of their competitors in (4) and (11) allows us to calculate the grammars that produce them. I present these below in the form of *skeletal bases* (Brasoveanu & Prince 2011). Each row shows the elementary ranking conditions (ERCs) that make up the grammars of the languages in question. Cells with ‘e’ are left blank for visual clarity. Column order is arbitrary and does not indicate ranking.

- (12) *Grammar for Group 1 phrasing (4a) in DRP+*

	M.XP	PRS	tH	A _t HR	DF	BM.φ	* _t	M.CP	NR
a.	W								L
b.		W						L	L
c.			W				W	L	
d.			W						L
e.				W					L
f.					W		W	L	
g.					W				L

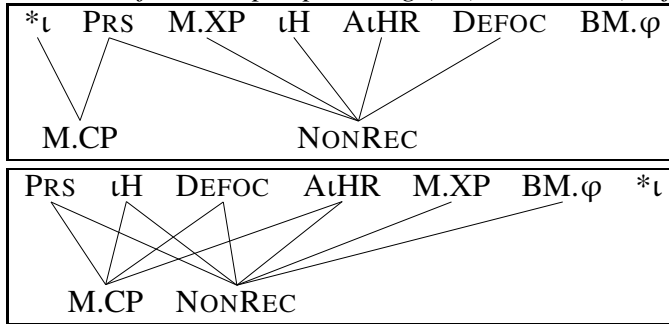
- (13) *Grammar for Group 2 phrasing (4b) in DRP+*

	PRS	tH	A _t HR	DF	NR	BM.φ	* _t	M.CP	M.XP
a.	W							L	L
b.		W					W	L	
c.		W							L
d.			W						L
e.				W			W	L	
f.				W					L
g.					W				L

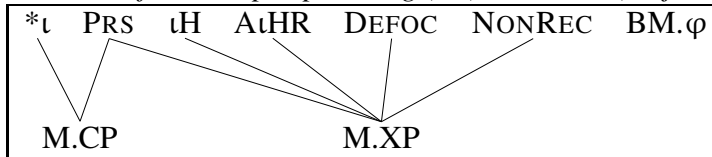
The skeletal basis is an ideal format for representing the grammars of these languages, because neither grammar is a partial order, as indicated by the presence of rows with multiple W's, indicating disjunctive domination. For example, row (12c), with ERC $\langle e, e, W, e, e, e, W, L, e \rangle$, indicates that in the grammar for the Group 1 phrasing (4a), at least one of tH or *_t dominates M.CP.

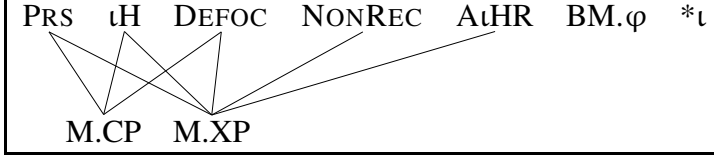
A less than ideal (though still accurate) way to represent grammars that are not partial orders is to use disjunctive Hasse diagrams (Prince 2017b). Using this method, the grammars in (12) and (13) can be represented as (14) and (15), respectively.

- (14) *Grammar for Group 1 phrasing (4a) in DRP+ (disjunctive Hasse representation of (12))*



- (15) *Grammar for Group 2 phrasing (4b) in DRP+ (disjunctive Hasse representation of (13))*





Some of the rankings in (12)/(14) and (13)/(15) accord with those proposed by Yip & Yuan (2025). For instance, for the Group 1 grammar, M.XP dominates NONREC, while the opposite holds for the Group 2 grammar (as seen in (12a) and (13g), respectively). Other rankings proposed by Yip & Yuan are only correct in DRP^+ insofar as they are portions of disjunctive ranking conditions. For instance, Yip & Yuan propose that DEFOCUS dominates M.CP for both Group 1 and Group 2, while in both grammars in DRP^+ , DEFOCUS *or* $*\iota$ dominates M.CP, as shown in rows (12f) and (13e).

Locally summing up, the problem of unwanted ties in DRP is resolved within a new system DRP^+ where I introduce two additional constraints: $*\iota$ and $\text{BINMAX}(\varphi, \text{branches})$. The needed addition of $*\iota$ partially invalidates Yip & Yuan’s ranking claims, though it does not result in the actual *reversal* of any of the rankings they propose.

3 Nonterminal correspondence and ν -heads

As mentioned in §2.1, Yip & Yuan (2025) make sporadic use of correspondence indices on non-terminal prosodic nodes, and these are relevant for the evaluation of DEFOCUS. Due to the sheer combinatorics of freely distributing correspondence indices on nonterminals, DRP and DRP^+ use a workaround that gives the same violation counts for DEFOCUS as seen in Yip & Yuan’s tableaux without reifying nonterminal-nonterminal correspondence.

In this section, I address a problem that emerges if an ι can count as a “correspondent” (in Yip & Yuan’s sense) of a defocused input constituent—though here too, no correspondence indices will be needed on prosodic nonterminals. To begin, consider Yip & Yuan’s definition of the DEFOCUS constraint:

(16) *Yip & Yuan’s (2025) definition of DEFOCUS* (p. 37, ex. (87))

Let Df be a defocus element and DfP be the highest prosodic constituent in the output corresponding to Df . Assign a violation mark if DfP is a prosodic head and a daughter of a higher prosodic category or a higher projection of the same category as DfP .

The evaluation of DEFOCUS hinges on the precise theory of input-output correspondence. In the candidates in their tableaux, Yip & Yuan use a subscript k to mark the output correspondent of the defocused input ZP . Whenever there is a φ with terminal string ω_{Z} , it is marked with k , and whenever there is no such φ , ω_{Z} is marked with k . Implicit here is an assumption that a φ or an ω may correspond with an input XP , but that an ι may not: no candidate is included showing correspondence $\text{ZP}_k \sim \iota_k$. This is why DEFOCUS is defined the way it is in (2f) in DRP and DRP^+ : by referring to the highest non- ι node in the output with terminal string ω_{Z} , we get the same violation counts as Yip & Yuan show in their tableaux.

However, the “non- ι ” condition in (2f) is a stipulation. There is no theory of why—in Yip & Yuan’s terms—the k marking the defocused output correspondent must be on a φ or ω . In DRP and DRP^+ , this is simply hard-coded into the definition of DEFOCUS to ensure the right

results. But suppose this assumption is lifted, and we assume that the highest node *of any prosodic category* with terminal string ω_Z is seen as the “correspondent” of Z’s phrase (as in the prosodic correspondence theory of Ito, Kitagawa & Mester 1996). This would mean redefining DEFOCUS more generally, without the “non- ι ” condition of (2f). I call this constraint DEFOCUS_{Any}, with “Any” alluding to the fact that *any* prosodic category can be the highest recognized as relevantly having terminal string ω_Z . Consider now a new system DRP.DFA, identical to DRP but with DEFOCUS_{Any} instead of DEFOCUS.

$$(17) \text{ DRP.DFA.GEN} \\ = \text{DRP.GEN (1)}$$

$$(18) \text{ DRP.DFA.CON} \\ \text{Constraints (2a–e) and (2g), along with the following constraint instead of (2f):}$$

- a. DEFOCUS_{Any} (DFA)
Violated once if the highest node in the output that shares a terminal string with a [defoc] input constituent is designated as a prosodic head.

Because no children of v are designated as heads in DRP.DFA.GEN, following the practice in Yip & Yuan’s candidate representations, an ι child of v with terminal string ω_Z is not a prosodic head, and candidates containing such an ι satisfy DEFOCUS_{Any}. As a result, DRP.DFA admits only one optimum, with perfect syntax-prosody matching. This optimum is shown in the following violation tableau to perfectly satisfy all constraints. All of its competitors (not shown here) fare worse on at least one constraint, meaning they are all harmonically bounded.

(19) Violation tableau containing sole optimum of DRP.DFA

[[CP [XP X] [YP Y]] [CP [ZP Z] _{defoc}]]		M.CP	M.XP	PRS	ι H	A ι HR	DFA	NR
a.		0	0	0	0	0	0	0

Matching of this type is exactly what Yip & Yuan set out to avoid. The problem arises because the highest prosodic node with terminal string ω_Z is the second ι , and this ι is not a prosodic head, since v has no head.

This problem can be avoided, and we can keep the more general definition of DEFOCUS_{Any}, by creating a system in which v can or must be headed. The most general approach would be to allow v to have zero or one prosodic head, in any position. The existence and position of v -heads would then need to be regulated by violable constraints. For the moment, however, I define a system with the strictest possible conditions on v -headedness, such that the rightmost child of v is obligatorily designated as a head. Making this one change to candidate representations but omitting $\ast\iota$ and BINMAX(ϕ ,branches) leads to a system with the same co-optimality problem as DRP, so I skip presenting a new system with this problem and jump straight to a system DRP. ι H+

with v -headedness, $\text{DEFocus}_{\text{Any}}$, $*\iota$, and $\text{BINMAX}(\varphi, \text{branches})$. DRP.vH+.GEN differs from DRP.GEN only on v -Headedness (20b-iv) as opposed to v -Unheadedness (1b-iv).

(20) DRP.vH+.GEN

- a. *Input*: $[[_{\text{CP}} [_{\text{XP}} \text{X}] [_{\text{YP}} \text{Y}]] [_{\text{CP}} [_{\text{ZP}} \text{Z}]_{\text{defoc}}]]$
 where the whole structure is a linearized syntactic tree rooted in an unlabeled constituent, with syntactic words X, Y, Z, syntactic phrases XP, YP, ZP, and clauses CP, with subscript “defoc” suffixed to a constituent bearing a [defocus] feature
- b. *Outputs*: An output is any prosodic tree P for which the following conditions hold:
 - i. *v-Rootedness*
 The root node of P is of category v .
 - ii. *v-Contains- ι*
 At least one child of the root v is of category ι .
 - iii. *v-Children*
 Every child of the root v is of category φ or ι .
 - iv. ***v-Headedness (the only difference between DRP.GEN and DRP.vH+.GEN)***
 The rightmost child of the root v is designated as the unique v -head.
 - v. *ι -Children*
 All children of ι nodes are of category φ .
 - vi. *Optional ι -Headedness*
 Zero or one child of every ι node is marked as that node’s head.
 - vii. *φ -Children*
 The children of a φ node are either all of category φ or all of category ω .
 - viii. *No Vacuous Recursion*
 If a φ is unary-branching, its sole child is of category ω .
 - ix. *φ -Headedness*
 Exactly one child of every φ node is marked as that node’s head.
 - x. *Terminal Correspondence*
 P has exactly three terminal nodes, and these are of category ω . They are ω_X , ω_Y , ω_Z , corresponding to the input words X, Y, Z, respectively.
 - xi. *Terminal Order*
 The linear order of P’s terminal nodes is $\omega_X \omega_Y \omega_Z$.

The full list of 180 candidates admitted by (20) is given in Appendix C.

The constraint set of DRP.vH+ is identical to that of DRP.DFA (18) but with the addition of $*\iota$ (8a) and $\text{BINMAX}(\varphi, \text{branches})$ (8b) from DRP+ :

(21) DRP.vH+.CON

- a. $\text{MATCH}(\text{CP}, \iota)$ (M.CP)
 Violated once by every input CP for which there is no matching output ι . (See (3) for the definition of “matching”).
- b. $\text{MATCH}(\text{XP}, \varphi)$ (M.XP)
 Violated once by every input XP for which there is no matching output φ . (An ‘XP’

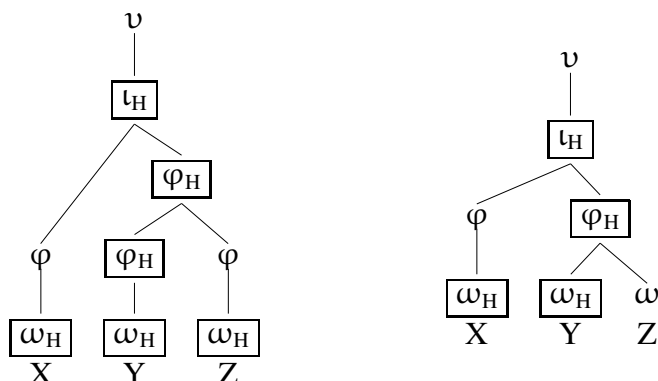
refers to any non-CP syntactic phrase, here including those labeled XP, YP, ZP. See (3) for the definition of “matching”.)

- c. PARSE (PRS)
Violated once by every φ that is immediately dominated by ν .
- d. ι HEAD (ι H)
Violated once by every ι that does not immediately dominate a node designated as a prosodic head.
- e. ALIGN- ι H-R (A ι HR)
Violated once by every prosodic head immediately dominated by an ι within which it is non-rightmost.
- f. DEFOCUS_{Any} (DEFOCUS_{Any}, DFA)
Violated once if the highest node in the output that shares a terminal string with a [defoc] input constituent is designated as a prosodic head.
- g. NONRECURSIVITY (NONREC, NR)
Violated once by every node of category κ that dominates another node of category κ .

The factorial typology of DRP.vH+ contains seven languages (as shown in (32) in Appendix A). Among them is one with the phrasing in (22a), which is a revised version of Yip & Yuan's Group 1 phrasing, and another with (22b), a revised version of their Group 2 phrasing. The only difference between the trees in (22) and (4) is that here the ι 's are v -heads.

(22) *Desired outputs with v-headedness*

- a. *Group 1 Phrasing (revised)* b. *Group 2 Phrasing (revised)*



These forms are able to emerge as optima in DRP.vH+ because they satisfy $\text{DEFOCUS}_{\text{Any}}$ and, given the headedness requirement on v , there is no candidate like (19a) (the sole optimum of DRP.DFA) admitted by DRP.vH+ .GEN. Instead, the closest counterpart to (19a) in the new system is one in which the ι with terminal string ω_Z is an v -head, causing the candidate to incur one violation of $\text{DEFOCUS}_{\text{Any}}$. This fact is recorded in the following violation tableau:

(23) *v-headed counterpart of (19a) in $DRP.vH+$*

[[_{CP} [_{XP} X] [_{YP} Y]] [_{CP} [_{ZP} Z] _{defoc}]]		M.CP	M.XP	PRS	ιH	AιHR	DFA	NR
a.		0	0	0	0	0	1	0

Now, instead of perfectly satisfying all constraints (like its counterpart (19a) in $DRP.DFA$), the right-headed matching candidate (23) does not harmonically bound all of its competitors and is open to competition, including by the desired optima in (22).

The grammars of the languages with Group 1 and Group 2 phrasings in $DRP.vH+$ are given in (24) and (25) as skeletal bases.

(24) *Grammar for Group 1 phrasing (22a) in $DRP.vH+$*

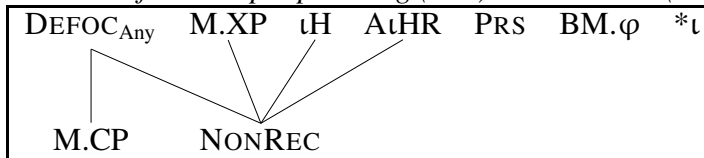
	M.XP	PRS	ιH	AιHR	DFA	BM.φ	*ι	M.CP	NONREC
a.	W								L
b.		W			W			L	
c.			W						L
d.				W					L
e.					W		W	L	
f.					W				L

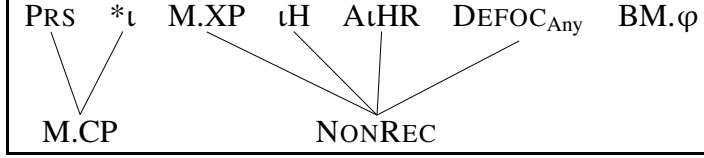
(25) *Grammar for Group 2 phrasing (22b) in $DRP.vH+$*

	PRS	ιH	AιHR	DFA	NONREC	BM.φ	*ι	M.CP	M.XP
a.	W			W				L	
b.		W							L
c.			W						L
d.				W			W	L	
e.				W					L
f.					W				L

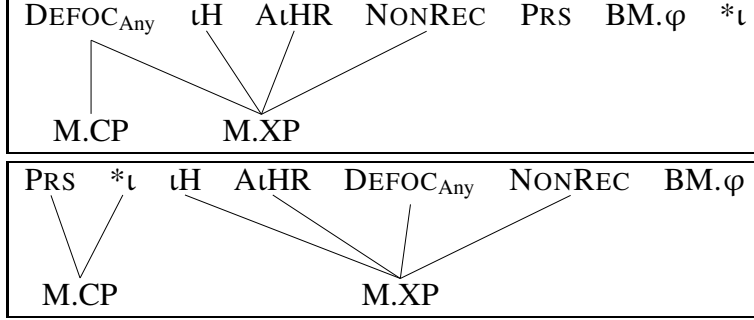
Just like the grammars of interest in $DRP+$, neither of these grammars is a partial order, but each can be represented by a pair of disjunctive Hasse diagrams. The double-Hasse representation of (24) is shown in (26), and that of (25) in (27).

(26) *Grammar for Group 1 phrasing (22a) in $DRP.vH+$ (disjunctive Hasse rep. of (24))*





(27) *Grammar for Group 2 phrasing (22b) in $DRP.vH+$ (disjunctive Hasse rep. of (25))*



While it expands on the original DRP in several ways, both in GEN (v -Headedness) and CON ($*\iota$, $BINMAX(\varphi, \text{branches})$), and the notion of “correspondence” baked into $DEFOCUS_{Any}$), the system $DRP.vH+$ preserves Yip & Yuan’s core insight that the need to “defocus” a constituent by denying it prosodic headship can drive syntax-prosody mismatches.

4 Conclusion

In conclusion, Yip & Yuan (2025) provide an intriguing proposal regarding the derivation of a defocusing-driven syntax-prosody mismatch, but their OT analysis requires certain amendments in order to work. I have explored their proposal by proceeding from OT system to OT system, following the method advocated by Prince (2017a,c) for inquiry within modern rigorous OT.

The first system presented here, DRP , adheres as closely to Yip & Yuan’s proposals as possible, as far as I have been able to determine. DRP goes some way toward deriving the desired outputs but falls short since these outputs tie with unwanted competitors with significantly different prosodic shapes. Next, I presented a second system, $DRP+$, which adds two standard prosodic markedness constraints: $*\iota$ and $BINMAX(\varphi, \text{branches})$. This was shown to eliminate the unwanted ties of DRP , yielding the correct outputs.

I then discussed a new system $DRP.DFA$ which removed a stipulation relevant to the evaluation of Yip & Yuan’s constraint $DEFOCUS$ from DRP and showed that on the most general view of “highest corresponding prosodic category”, a system without v -heads $DRP.DFA$ is entirely unable to derive the desired syntax-prosody mismatches. Modifying this system to require v -headedness at the right edge and adding $*\iota$ and $BINMAX(\varphi, \text{branches})$ produces a final system, $DRP.vH+$, which yields the correct outputs without the stipulations on $DEFOCUS$ /correspondence of DRP and $DRP+$.

These remarks highlight the importance of defining OT systems and taking all candidates and constraint violations into account, as has been recognized in much recent work on syntax-prosody mapping in OT (Bellik & Kalivoda 2016, 2019; Kalivoda 2018, 2022; Van Handel 2021; Bellik, Ito, Kalivoda & Mester 2022; Ishihara & Kalivoda 2022; Alber & Prince 2022; Elkins 2023; and papers in Bellik, Ito, Kalivoda & Mester 2023).

Finally, the systems discussed above are quite limited in scope, addressing only the prosodic parsing of one syntactic input. There is no guarantee that the grammars that succeed in generating the desired optima for that input would generate desired optima for others. The way forward, then, is to continue defining and testing similar systems with more syntactic inputs—the most reasonable place to begin being with other short and simple sentences, including some without defocusing. Following this method will reveal whether the type of analysis proposed by Yip & Yuan (2025) and refined here is ultimately viable.

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Appendix A: Factorial typologies

- (28) *Notation for bracketed string representations of outputs*
- $\langle_v \rangle$ utterance edges
 - $\{\iota\}$ intonational phrase edges
 - (φ) phonological phrase edges
 - ω_α prosodic word with input correspondent α
 - $'$ prosodic head marker (suffixal)

(29) *Factorial typology of DRP (GEN in (1), CON in (2))*⁹

	[[CP [XP X] [YP Y]] [CP [ZP Z] _{defoc}]]
L.1	$\langle \nu \{ \iota (\varphi \omega_X')' \} \{ \iota (\varphi \omega_Y' \omega_Z')' \} \rangle$ $\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y' \omega_Z')' \} \rangle \quad \leftarrow \text{Group 2 desired phrasing}$
L.2	$\langle \nu \{ \iota (\varphi \omega_X')' \} \{ \iota (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$ $\langle \nu \{ \iota (\varphi (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$ $\langle \nu \{ \iota (\varphi (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z'))' \} \rangle$ $\langle \nu \{ \iota (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle \quad \leftarrow \text{Group 1 desired phrasing}$
L.3	$\langle \nu \{ \iota (\varphi \omega_X')' \} \{ \iota (\varphi \omega_Y')' (\varphi \omega_Z') \} \rangle$ $\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z') \} \rangle$ $\langle \nu \{ \iota (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$
L.4	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} (\varphi \omega_Z') \rangle$
L.5	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} \{ \iota (\varphi \omega_Z') \} \rangle$
L.6	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} \{ \iota (\varphi \omega_Z')' \} \rangle$

(30) *Factorial typology of DRP+ (GEN in (7), CON in (8))*

	[[CP [XP X] [YP Y]] [CP [ZP Z] _{defoc}]]
L.1	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} (\varphi \omega_Z') \rangle$
L.2	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} \{ \iota (\varphi \omega_Z') \} \rangle$
L.3	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} \{ \iota (\varphi \omega_Z')' \} \rangle$
L.4	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y' \omega_Z')' \} \rangle \quad \leftarrow \text{Group 2 desired phrasing}$
L.5	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle \quad \leftarrow \text{Group 1 desired phrasing}$
L.6	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$
L.7	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$
L.8	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z') \} \rangle$ $\langle \nu \{ \iota (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$

(31) *Factorial typology of DRP.DFA (GEN in (17), CON in (18))*

	[[CP [XP X] [YP Y]] [CP [ZP Z] _{defoc}]]
L.1	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} \{ \iota (\varphi \omega_Z')' \} \rangle \quad \leftarrow \text{Sole winner (match)}$

(32) *Factorial typology of DRP.vH+ (GEN in (20), CON in (21))*

	[[CP [XP X] [YP Y]] [CP [ZP Z] _{defoc}]]
L.1	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} (\varphi \omega_Z')' \rangle$
L.2	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' \} \{ \iota (\varphi \omega_Z')' \} \rangle$
L.3	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y' \omega_Z')' \} \rangle \quad \leftarrow \text{Group 2 desired winner}$
L.4	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle \quad \leftarrow \text{Group 1 desired winner}$
L.5	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$
L.6	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$
L.7	$\langle \nu \{ \iota (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z')' \} \rangle$ $\langle \nu \{ \iota (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$

⁹Outputs in the same cell are co-optima.

Appendix B: Output trees for DRP, DRP+, and DRP.DFA

1. $\langle_v (\varphi \omega_{X'}) (\varphi \omega_{Y'}) \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
2. $\langle_v (\varphi \omega_{X'}) (\varphi \omega_{Y'}) \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
3. $\langle_v (\varphi \omega_{X'}) \{_{\iota} (\varphi \omega_{Y'})\} (\varphi \omega_{Z'})\rangle$
4. $\langle_v (\varphi \omega_{X'}) \{_{\iota} (\varphi \omega_{Y'})\} \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
5. $\langle_v (\varphi \omega_{X'}) \{_{\iota} (\varphi \omega_{Y'})\} \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
6. $\langle_v (\varphi \omega_{X'}) \{_{\iota} (\varphi \omega_{Y'})'\} (\varphi \omega_{Z'})\rangle$
7. $\langle_v (\varphi \omega_{X'}) \{_{\iota} (\varphi \omega_{Y'})'\} \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
8. $\langle_v (\varphi \omega_{X'}) \{_{\iota} (\varphi \omega_{Y'})'\} \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
9. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} (\varphi \omega_{Y'}) (\varphi \omega_{Z'})\rangle$
10. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} (\varphi \omega_{Y'}) \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
11. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} (\varphi \omega_{Y'}) \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
12. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} \{_{\iota} (\varphi \omega_{Y'})\} (\varphi \omega_{Z'})\rangle$
13. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} \{_{\iota} (\varphi \omega_{Y'})\} \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
14. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} \{_{\iota} (\varphi \omega_{Y'})\} \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
15. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} \{_{\iota} (\varphi \omega_{Y'})'\} (\varphi \omega_{Z'})\rangle$
16. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} \{_{\iota} (\varphi \omega_{Y'})'\} \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
17. $\langle_v \{_{\iota} (\varphi \omega_{X'})\} \{_{\iota} (\varphi \omega_{Y'})'\} \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
18. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} (\varphi \omega_{Y'}) (\varphi \omega_{Z'})\rangle$
19. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} (\varphi \omega_{Y'}) \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
20. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} (\varphi \omega_{Y'}) \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
21. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} \{_{\iota} (\varphi \omega_{Y'})\} (\varphi \omega_{Z'})\rangle$
22. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} \{_{\iota} (\varphi \omega_{Y'})\} \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
23. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} \{_{\iota} (\varphi \omega_{Y'})\} \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$
24. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} \{_{\iota} (\varphi \omega_{Y'})'\} (\varphi \omega_{Z'})\rangle$
25. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} \{_{\iota} (\varphi \omega_{Y'})'\} \{_{\iota} (\varphi \omega_{Z'})\}\rangle$
26. $\langle_v \{_{\iota} (\varphi \omega_{X'})'\} \{_{\iota} (\varphi \omega_{Y'})'\} \{_{\iota} (\varphi \omega_{Z'})'\}\rangle$

27. $\langle_v (\varphi \omega_X') \{ \iota (\varphi \omega_Y \omega_Z') \} \rangle$
28. $\langle_v (\varphi \omega_X') \{ \iota (\varphi \omega_Y' \omega_Z) \} \rangle$
29. $\langle_v (\varphi \omega_X') \{ \iota (\varphi \omega_Y \omega_Z')' \} \rangle$
30. $\langle_v (\varphi \omega_X') \{ \iota (\varphi \omega_Y' \omega_Z')' \} \rangle$
31. $\langle_v (\varphi \omega_X') \{ \iota (\varphi (\varphi \omega_Y') (\varphi \omega_Z')') \} \rangle$
32. $\langle_v (\varphi \omega_X') \{ \iota (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \} \rangle$
33. $\langle_v (\varphi \omega_X') \{ \iota (\varphi (\varphi \omega_Y') (\varphi \omega_Z')')' \} \rangle$
34. $\langle_v (\varphi \omega_X') \{ \iota (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$
35. $\langle_v (\varphi \omega_X') \{ \iota (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$
36. $\langle_v (\varphi \omega_X') \{ \iota (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$
37. $\langle_v (\varphi \omega_X') \{ \iota (\varphi \omega_Y')' (\varphi \omega_Z') \} \rangle$
38. $\langle_v \{ \iota (\varphi \omega_X') \} (\varphi \omega_Y \omega_Z') \rangle$
39. $\langle_v \{ \iota (\varphi \omega_X') \} (\varphi \omega_Y' \omega_Z) \rangle$
40. $\langle_v \{ \iota (\varphi \omega_X') \} (\varphi (\varphi \omega_Y') (\varphi \omega_Z')') \rangle$
41. $\langle_v \{ \iota (\varphi \omega_X') \} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \rangle$
42. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi \omega_Y \omega_Z') \} \rangle$
43. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi \omega_Y' \omega_Z) \} \rangle$
44. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi \omega_Y \omega_Z')' \} \rangle$
45. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi \omega_Y' \omega_Z')' \} \rangle$
46. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi (\varphi \omega_Y') (\varphi \omega_Z')') \} \rangle$
47. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \} \rangle$
48. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi (\varphi \omega_Y') (\varphi \omega_Z')')' \} \rangle$
49. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$
50. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$
51. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$
52. $\langle_v \{ \iota (\varphi \omega_X') \} \{ \iota (\varphi \omega_Y')' (\varphi \omega_Z') \} \rangle$
53. $\langle_v \{ \iota (\varphi \omega_X')' \} (\varphi \omega_Y \omega_Z') \rangle$

54. $\langle_v \{_{\iota} (\varphi \omega_X')' \} (\varphi \omega_Y' \omega_Z) \rangle$
55. $\langle_v \{_{\iota} (\varphi \omega_X')' \} (\varphi (\varphi \omega_Y') (\varphi \omega_Z')) \rangle$
56. $\langle_v \{_{\iota} (\varphi \omega_X')' \} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \rangle$
57. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y' \omega_Z') \} \rangle$
58. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y' \omega_Z') \} \rangle$
59. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y' \omega_Z')' \} \rangle$
60. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y' \omega_Z')' \} \rangle$
61. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z')) \} \rangle$
62. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \} \rangle$
63. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \} \rangle$
64. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$
65. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$
66. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$
67. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y')' (\varphi \omega_Z') \} \rangle$
68. $\langle_v (\varphi \omega_X \omega_Y') \{_{\iota} (\varphi \omega_Z') \} \rangle$
69. $\langle_v (\varphi \omega_X \omega_Y') \{_{\iota} (\varphi \omega_Z')' \} \rangle$
70. $\langle_v (\varphi \omega_X' \omega_Y) \{_{\iota} (\varphi \omega_Z') \} \rangle$
71. $\langle_v (\varphi \omega_X' \omega_Y) \{_{\iota} (\varphi \omega_Z')' \} \rangle$
72. $\langle_v (\varphi (\varphi \omega_X') (\varphi \omega_Y')) \{_{\iota} (\varphi \omega_Z') \} \rangle$
73. $\langle_v (\varphi (\varphi \omega_X') (\varphi \omega_Y')) \{_{\iota} (\varphi \omega_Z')' \} \rangle$
74. $\langle_v (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) \{_{\iota} (\varphi \omega_Z') \} \rangle$
75. $\langle_v (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) \{_{\iota} (\varphi \omega_Z')' \} \rangle$
76. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') \} (\varphi \omega_Z') \rangle$
77. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') \} \{_{\iota} (\varphi \omega_Z') \} \rangle$
78. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \} \rangle$
79. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) \} (\varphi \omega_Z') \rangle$
80. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) \} \{_{\iota} (\varphi \omega_Z') \} \rangle$

81. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y)\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
82. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')'\} (\varphi \omega_Z')\rangle$
83. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')'\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
84. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')'\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
85. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y')'\} (\varphi \omega_Z')\rangle$
86. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y')'\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
87. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y')'\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
88. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')\} (\varphi \omega_Z')\rangle$
89. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
90. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
91. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))\} (\varphi \omega_Z')\rangle$
92. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
93. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
94. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')'\} (\varphi \omega_Z')\rangle$
95. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')'\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
96. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')'\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
97. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))'\} (\varphi \omega_Z')\rangle$
98. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))'\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
99. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))'\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
100. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')\} (\varphi \omega_Z')\rangle$
101. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
102. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
103. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')'\} (\varphi \omega_Z')\rangle$
104. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')'\} \{_{\iota} (\varphi \omega_Z')\}\rangle$
105. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')'\} \{_{\iota} (\varphi \omega_Z')'\}\rangle$
106. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y')\} (\varphi \omega_Z')\rangle$
107. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y')\} \{_{\iota} (\varphi \omega_Z')\}\rangle$

108. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \} \rangle$
109. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y \omega_Z') \} \rangle$
110. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y' \omega_Z) \} \rangle$
111. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y \omega_Z) \} \rangle$
112. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y \omega_Z')' \} \rangle$
113. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y' \omega_Z')' \} \rangle$
114. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y \omega_Z')' \} \rangle$
115. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y') (\varphi \omega_Z')') \} \rangle$
116. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y) (\varphi \omega_Z')') \} \rangle$
117. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y')' (\varphi \omega_Z')) \} \rangle$
118. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y)' (\varphi \omega_Z')) \} \rangle$
119. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y') (\varphi \omega_Z')')' \} \rangle$
120. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y) (\varphi \omega_Z')')' \} \rangle$
121. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y')' (\varphi \omega_Z'))' \} \rangle$
122. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y)' (\varphi \omega_Z'))' \} \rangle$
123. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')') (\varphi \omega_Z')') \} \rangle$
124. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) (\varphi \omega_Z')') \} \rangle$
125. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' (\varphi \omega_Z')) \} \rangle$
126. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' (\varphi \omega_Z')) \} \rangle$
127. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')') (\varphi \omega_Z')')' \} \rangle$
128. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) (\varphi \omega_Z')')' \} \rangle$
129. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' (\varphi \omega_Z'))' \} \rangle$
130. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' (\varphi \omega_Z'))' \} \rangle$
131. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y \omega_Z')') \} \rangle$
132. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y' \omega_Z')') \} \rangle$
133. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y \omega_Z')) \} \rangle$
134. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y' \omega_Z)) \} \rangle$

135. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y \omega_Z')')' \} \rangle$
136. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y' \omega_Z')')' \} \rangle$
137. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y \omega_Z'))' \} \rangle$
138. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y' \omega_Z))' \} \rangle$
139. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z')'))' \} \rangle$
140. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))')' \} \rangle$
141. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y') (\varphi \omega_Z')'))' \} \rangle$
142. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))))' \} \rangle$
143. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z')'))' \} \rangle$
144. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))')' \} \rangle$
145. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y') (\varphi \omega_Z')'))' \} \rangle$
146. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))))' \} \rangle$
147. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z'))' \} \rangle$
148. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$
149. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z'))' \} \rangle$
150. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z')')' \} \rangle$
151. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$
152. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z'))' \} \rangle$
153. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') (\varphi \omega_Z')' \} \rangle$
154. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) (\varphi \omega_Z')' \} \rangle$
155. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') (\varphi \omega_Z')' \} \rangle$
156. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) (\varphi \omega_Z')' \} \rangle$
157. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')' (\varphi \omega_Z')' \} \rangle$
158. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y)' (\varphi \omega_Z')' \} \rangle$
159. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')') (\varphi \omega_Z')' \} \rangle$
160. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) (\varphi \omega_Z')' \} \rangle$
161. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')') (\varphi \omega_Z')' \} \rangle$

162. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')) (\varphi \omega_Z')' \} \rangle$
163. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' (\varphi \omega_Z') \} \rangle$
164. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y'))' (\varphi \omega_Z') \} \rangle$
165. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y \omega_Z') \} \rangle$
166. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y' \omega_Z) \} \rangle$
167. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y \omega_Z')' \} \rangle$
168. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y' \omega_Z) \} \rangle$
169. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y \omega_Z') \} \rangle$
170. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y' \omega_Z) \} \rangle$
171. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z')') \} \rangle$
172. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \} \rangle$
173. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z')')' \} \rangle$
174. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \} \rangle$
175. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi (\varphi \omega_Y') (\varphi \omega_Z')') \} \rangle$
176. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \} \rangle$
177. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$
178. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z')' \} \rangle$
179. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z') \} \rangle$
180. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z') \} \rangle$

Appendix C: Output trees for DRP.vH+

1. $\langle_v (\varphi \omega_X') (\varphi \omega_Y') \{_{\iota} (\varphi \omega_Z')' \} \rangle$
2. $\langle_v (\varphi \omega_X') (\varphi \omega_Y') \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
3. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi \omega_Y') \} (\varphi \omega_Z')' \rangle$
4. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \} \rangle$
5. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
6. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi \omega_Y')' \} (\varphi \omega_Z')' \rangle$

7. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi \omega_Y')' \} \{ {}_l (\varphi \omega_Z') \}' \rangle$
8. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi \omega_Y')' \} \{ {}_l (\varphi \omega_Z')' \}' \rangle$
9. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} (\varphi \omega_Y') (\varphi \omega_Z')' \rangle$
10. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} (\varphi \omega_Y') \{ {}_l (\varphi \omega_Z') \}' \rangle$
11. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} (\varphi \omega_Y') \{ {}_l (\varphi \omega_Z')' \}' \rangle$
12. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} \{ {}_l (\varphi \omega_Y') \} (\varphi \omega_Z')' \rangle$
13. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} \{ {}_l (\varphi \omega_Y') \} \{ {}_l (\varphi \omega_Z') \}' \rangle$
14. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} \{ {}_l (\varphi \omega_Y') \} \{ {}_l (\varphi \omega_Z')' \}' \rangle$
15. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} \{ {}_l (\varphi \omega_Y')' \} (\varphi \omega_Z')' \rangle$
16. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} \{ {}_l (\varphi \omega_Y')' \} \{ {}_l (\varphi \omega_Z') \}' \rangle$
17. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X') \} \{ {}_l (\varphi \omega_Y')' \} \{ {}_l (\varphi \omega_Z')' \}' \rangle$
18. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} (\varphi \omega_Y') (\varphi \omega_Z')' \rangle$
19. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} (\varphi \omega_Y') \{ {}_l (\varphi \omega_Z') \}' \rangle$
20. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} (\varphi \omega_Y') \{ {}_l (\varphi \omega_Z')' \}' \rangle$
21. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} \{ {}_l (\varphi \omega_Y') \} (\varphi \omega_Z')' \rangle$
22. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} \{ {}_l (\varphi \omega_Y') \} \{ {}_l (\varphi \omega_Z') \}' \rangle$
23. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} \{ {}_l (\varphi \omega_Y') \} \{ {}_l (\varphi \omega_Z')' \}' \rangle$
24. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} \{ {}_l (\varphi \omega_Y')' \} (\varphi \omega_Z')' \rangle$
25. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} \{ {}_l (\varphi \omega_Y')' \} \{ {}_l (\varphi \omega_Z') \}' \rangle$
26. $\langle_{\text{v}} \{ {}_l (\varphi \omega_X')' \} \{ {}_l (\varphi \omega_Y')' \} \{ {}_l (\varphi \omega_Z')' \}' \rangle$
27. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi \omega_Y \omega_Z') \}' \rangle$
28. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi \omega_Y' \omega_Z) \}' \rangle$
29. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi \omega_Y \omega_Z')' \}' \rangle$
30. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi \omega_Y' \omega_Z)' \}' \rangle$
31. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
32. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi (\varphi \omega_Y')' (\varphi \omega_Z)))' \}' \rangle$
33. $\langle_{\text{v}} (\varphi \omega_X') \{ {}_l (\varphi (\varphi \omega_Y') (\varphi \omega_Z')')' \}' \rangle$

34. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \}' \rangle$
35. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z') \}' \rangle$
36. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z')' \}' \rangle$
37. $\langle_v (\varphi \omega_X') \{_{\iota} (\varphi \omega_Y')' (\varphi \omega_Z') \}' \rangle$
38. $\langle_v \{_{\iota} (\varphi \omega_X') \} (\varphi \omega_Y \omega_Z')' \rangle$
39. $\langle_v \{_{\iota} (\varphi \omega_X') \} (\varphi \omega_Y' \omega_Z')' \rangle$
40. $\langle_v \{_{\iota} (\varphi \omega_X') \} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \rangle$
41. $\langle_v \{_{\iota} (\varphi \omega_X') \} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \rangle$
42. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi \omega_Y \omega_Z') \}' \rangle$
43. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi \omega_Y' \omega_Z') \}' \rangle$
44. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi \omega_Y \omega_Z')' \}' \rangle$
45. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi \omega_Y' \omega_Z')' \}' \rangle$
46. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z')) \}' \rangle$
47. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \}' \rangle$
48. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
49. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \}' \rangle$
50. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z') \}' \rangle$
51. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z')' \}' \rangle$
52. $\langle_v \{_{\iota} (\varphi \omega_X') \} \{_{\iota} (\varphi \omega_Y')' (\varphi \omega_Z') \}' \rangle$
53. $\langle_v \{_{\iota} (\varphi \omega_X')' \} (\varphi \omega_Y \omega_Z')' \rangle$
54. $\langle_v \{_{\iota} (\varphi \omega_X')' \} (\varphi \omega_Y' \omega_Z')' \rangle$
55. $\langle_v \{_{\iota} (\varphi \omega_X')' \} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \rangle$
56. $\langle_v \{_{\iota} (\varphi \omega_X')' \} (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \rangle$
57. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y \omega_Z') \}' \rangle$
58. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y' \omega_Z') \}' \rangle$
59. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y \omega_Z')' \}' \rangle$
60. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y' \omega_Z')' \}' \rangle$

61. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
62. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
63. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
64. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
65. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z') \}' \rangle$
66. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y') (\varphi \omega_Z')' \}' \rangle$
67. $\langle_v \{_{\iota} (\varphi \omega_X')' \} \{_{\iota} (\varphi \omega_Y')' (\varphi \omega_Z')' \}' \rangle$
68. $\langle_v (\varphi \omega_X \omega_Y') \{_{\iota} (\varphi \omega_Z') \}' \rangle$
69. $\langle_v (\varphi \omega_X \omega_Y') \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
70. $\langle_v (\varphi \omega_X' \omega_Y) \{_{\iota} (\varphi \omega_Z') \}' \rangle$
71. $\langle_v (\varphi \omega_X' \omega_Y) \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
72. $\langle_v (\varphi (\varphi \omega_X') (\varphi \omega_Y'))' \{_{\iota} (\varphi \omega_Z') \}' \rangle$
73. $\langle_v (\varphi (\varphi \omega_X') (\varphi \omega_Y'))' \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
74. $\langle_v (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) \{_{\iota} (\varphi \omega_Z') \}' \rangle$
75. $\langle_v (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
76. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') \} (\varphi \omega_Z')' \rangle$
77. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
78. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
79. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) \} (\varphi \omega_Z')' \rangle$
80. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
81. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
82. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')' \} (\varphi \omega_Z')' \rangle$
83. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')' \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
84. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')' \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
85. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y') \} (\varphi \omega_Z')' \rangle$
86. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y') \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
87. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$

88. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')) \} (\varphi \omega_Z')' \rangle$
89. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')) \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
90. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')) \} \{_{\iota} (\varphi \omega_Z') \}'' \rangle$
91. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) \} (\varphi \omega_Z')' \rangle$
92. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
93. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) \} \{_{\iota} (\varphi \omega_Z') \}'' \rangle$
94. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' \} (\varphi \omega_Z')' \rangle$
95. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
96. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
97. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' \} (\varphi \omega_Z')' \rangle$
98. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
99. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
100. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y') \} (\varphi \omega_Z')' \rangle$
101. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y') \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
102. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
103. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')' \} (\varphi \omega_Z')' \rangle$
104. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')' \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
105. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')' \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
106. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y') \} (\varphi \omega_Z')' \rangle$
107. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y') \} \{_{\iota} (\varphi \omega_Z') \}' \rangle$
108. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y') \} \{_{\iota} (\varphi \omega_Z')' \}' \rangle$
109. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y \omega_Z') \}' \rangle$
110. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y' \omega_Z) \}' \rangle$
111. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y \omega_Z) \}' \rangle$
112. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y \omega_Z')' \}' \rangle$
113. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y' \omega_Z)' \}' \rangle$
114. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y \omega_Z)' \}' \rangle$

115. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
116. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y) (\varphi \omega_Z'))' \}' \rangle$
117. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y')' (\varphi \omega_Z'))' \}' \rangle$
118. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y)' (\varphi \omega_Z'))' \}' \rangle$
119. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y') (\varphi \omega_Z')')' \}' \rangle$
120. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y) (\varphi \omega_Z')')' \}' \rangle$
121. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X \omega_Y')' (\varphi \omega_Z'))' \}' \rangle$
122. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X' \omega_Y)' (\varphi \omega_Z'))' \}' \rangle$
123. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')') (\varphi \omega_Z')')' \}' \rangle$
124. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) (\varphi \omega_Z')')' \}' \rangle$
125. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' (\varphi \omega_Z'))' \}' \rangle$
126. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' (\varphi \omega_Z'))' \}' \rangle$
127. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')') (\varphi \omega_Z')')' \}' \rangle$
128. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) (\varphi \omega_Z')')' \}' \rangle$
129. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X') (\varphi \omega_Y')')' (\varphi \omega_Z'))' \}' \rangle$
130. $\langle_v \{_{\iota} (\varphi (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' (\varphi \omega_Z'))' \}' \rangle$
131. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y \omega_Z'))' \}' \rangle$
132. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y' \omega_Z'))' \}' \rangle$
133. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y \omega_Z'))' \}' \rangle$
134. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y' \omega_Z'))' \}' \rangle$
135. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y \omega_Z')')' \}' \rangle$
136. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y' \omega_Z')')' \}' \rangle$
137. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y \omega_Z'))' \}' \rangle$
138. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y' \omega_Z'))' \}' \rangle$
139. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z')'))' \}' \rangle$
140. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))')' \}' \rangle$
141. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))')' \}' \rangle$

142. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))))\}' \rangle$
143. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))')\}' \rangle$
144. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))')\}' \rangle$
145. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))')\}' \rangle$
146. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))')\}' \rangle$
147. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z'))\}' \rangle$
148. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z'))\}' \rangle$
149. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z'))\}' \rangle$
150. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z'))'\}' \rangle$
151. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z'))'\}' \rangle$
152. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z'))'\}' \rangle$
153. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') (\varphi \omega_Z')\}' \rangle$
154. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) (\varphi \omega_Z')\}' \rangle$
155. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y') (\varphi \omega_Z')\}' \rangle$
156. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y) (\varphi \omega_Z')\}' \rangle$
157. $\langle_v \{_{\iota} (\varphi \omega_X \omega_Y')' (\varphi \omega_Z')\}' \rangle$
158. $\langle_v \{_{\iota} (\varphi \omega_X' \omega_Y)' (\varphi \omega_Z')\}' \rangle$
159. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')) (\varphi \omega_Z')\}' \rangle$
160. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) (\varphi \omega_Z')\}' \rangle$
161. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y')) (\varphi \omega_Z')'\}' \rangle$
162. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y')) (\varphi \omega_Z')'\}' \rangle$
163. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X') (\varphi \omega_Y'))' (\varphi \omega_Z')\}' \rangle$
164. $\langle_v \{_{\iota} (\varphi (\varphi \omega_X')' (\varphi \omega_Y'))' (\varphi \omega_Z')\}' \rangle$
165. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y \omega_Z')\}' \rangle$
166. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y' \omega_Z)\}' \rangle$
167. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y \omega_Z')\}' \rangle$
168. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y' \omega_Z)\}' \rangle$

169. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y \omega_Z') \}' \rangle$
170. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y' \omega_Z) \}' \rangle$
171. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z')) \}' \rangle$
172. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \}' \rangle$
173. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y') (\varphi \omega_Z'))' \}' \rangle$
174. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi (\varphi \omega_Y')' (\varphi \omega_Z'))' \}' \rangle$
175. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi (\varphi \omega_Y') (\varphi \omega_Z')) \}' \rangle$
176. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi (\varphi \omega_Y')' (\varphi \omega_Z')) \}' \rangle$
177. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z') \}' \rangle$
178. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y') (\varphi \omega_Z')' \}' \rangle$
179. $\langle_v \{_{\iota} (\varphi \omega_X') (\varphi \omega_Y')' (\varphi \omega_Z') \}' \rangle$
180. $\langle_v \{_{\iota} (\varphi \omega_X')' (\varphi \omega_Y') (\varphi \omega_Z') \}' \rangle$